

Can PR-Level Features Explain and Predict GitHub PR Merge Outcomes?

Answer: yes, but only as moderate predictive association.

- Headline model: **Random forest balanced**
- Feature contract: **25 leakage-safer PR-level features**
- Official test not-merged F1: **0.314** default, **0.326** validation-tuned
- Review-process contract F1: **0.436**, showing later process signal
- Claim boundary: **not causal, not deployment-ready**, weaker under stricter generalization

Research Design

Decision	Final choice
Question	Can PR-level features explain and predict merge outcomes?
Target	<code>merged_or_not</code>
Main data	PRFeatures train/test files
Headline claim	Early/submission-time feature association
Main risk	Feature timing and project/creator overlap

The project is framed as an empirical data-analysis study, not a merge-decision product.

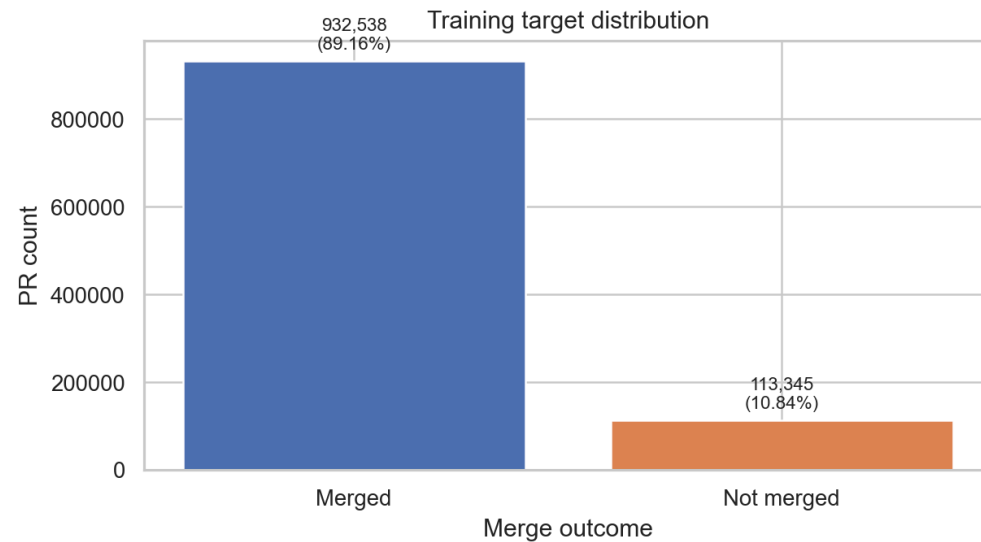
Dataset and Scope

Source: Zenodo PRFeatures dataset from *GitHub Pull Request Analysis: Sentiment Data and Developer Survey Responses*.

File	Role	Rows	Columns
prfeatures_train_data.csv	model development / final training	1,045,883	72
prfeatures_test_data.csv	untouched official test	260,195	72
pr_comments_dataset_publish.csv	profiled, not joined	588,097	15
survey_responses_raw.csv	documented, not main model source	22	56

The comment file lacks the numeric PRFeatures join keys, so it is profiled rather than force-joined.

Target and Data Characteristics



Split	Merged	Not merged
Train	932,538 (89.16%)	113,345 (10.84%)
Test	232,073 (89.19%)	28,122 (10.81%)

Accuracy alone is misleading: a majority baseline has 89% accuracy and 0 not-merged recall.

Feature-Availability Contract

Feature group	Treatment	Reason
Direct identifiers	Excluded	identity can memorize ecosystems
Post-outcome fields	Excluded	close-time and lifetime variables leak outcome process
Comments/sentiment/CI outcomes	Excluded	can encode review process after submission
prior_review_num	Sensitivity only	reviewer/integrator-context assumption
Headline features	Used	contributor history, project context, PR scope, testing context

This is a defensible early-information model, not the highest possible metric model.

Professor Defense Spine

Likely challenge	Answer we defend
Are you forcing a result?	No. The headline result is moderate, and stronger T2 results are labeled later-stage.
Is there leakage?	Direct ids, post-outcome fields, comments, CI outcomes, and risky success-rate fields are excluded or sensitivity-only.
Does the math make sense?	Metrics are confusion-matrix based, validation-selected, and checked against the untouched official test.
Does it generalize?	Partially. Official PR ids are clean, but project/creator overlap limits external validity.
What is the answer?	PR-level signal exists, but prediction strength depends on feature availability.

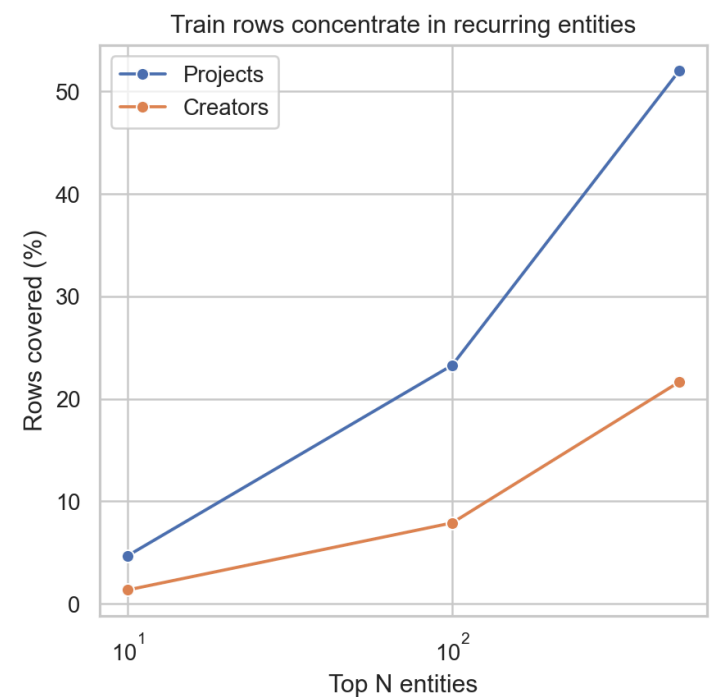
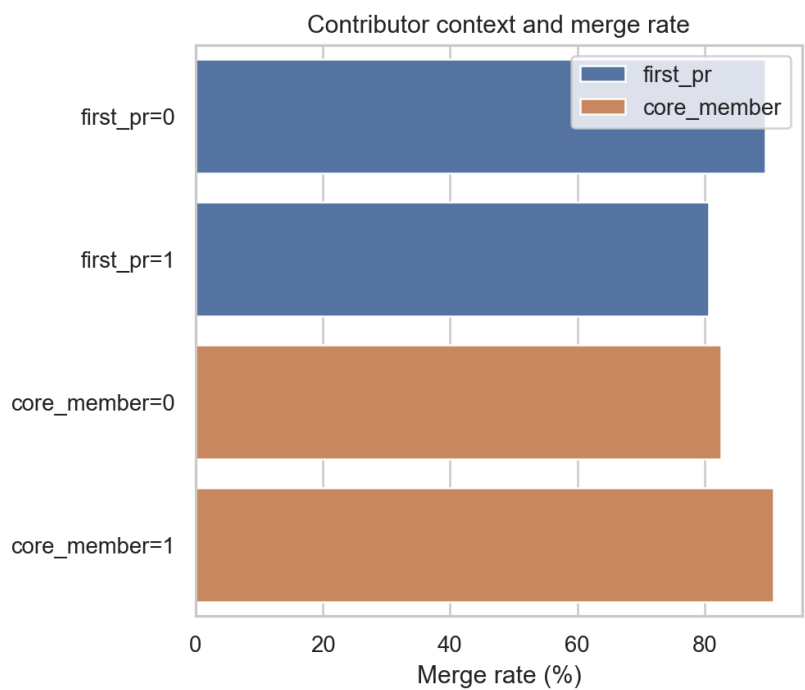
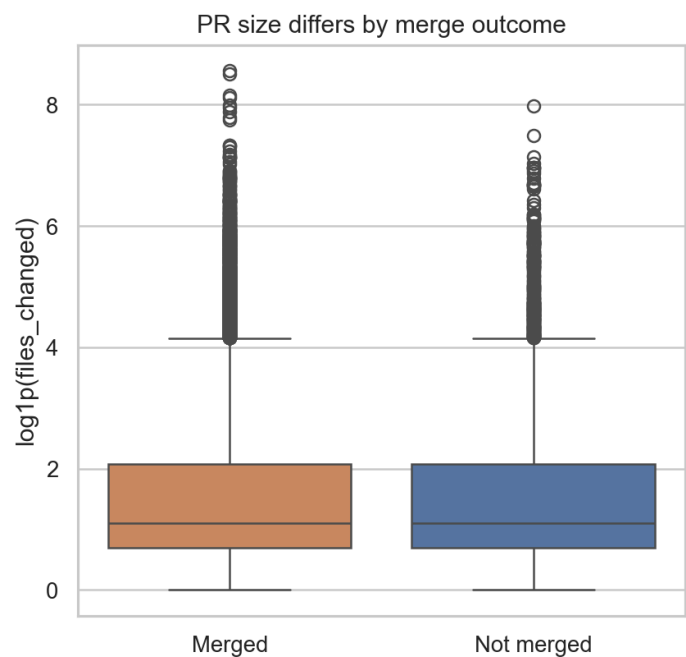
This is the line to hold in Q&A.

EDA Findings

Finding	Evidence
First PRs are riskier	19.32% not merged vs 10.55% for non-first PRs
Core members differ	9.21% not merged vs 17.44% for non-core contributors
Prior project history matters	median previous PRs: 22 not-merged vs 41 merged
Project concentration is high	top 500 projects contain 52.03% of training rows
CI presence differs	9.91% not merged with CI vs 12.95% without CI

These are associations in observational data, not causal effects.

EDA Evidence



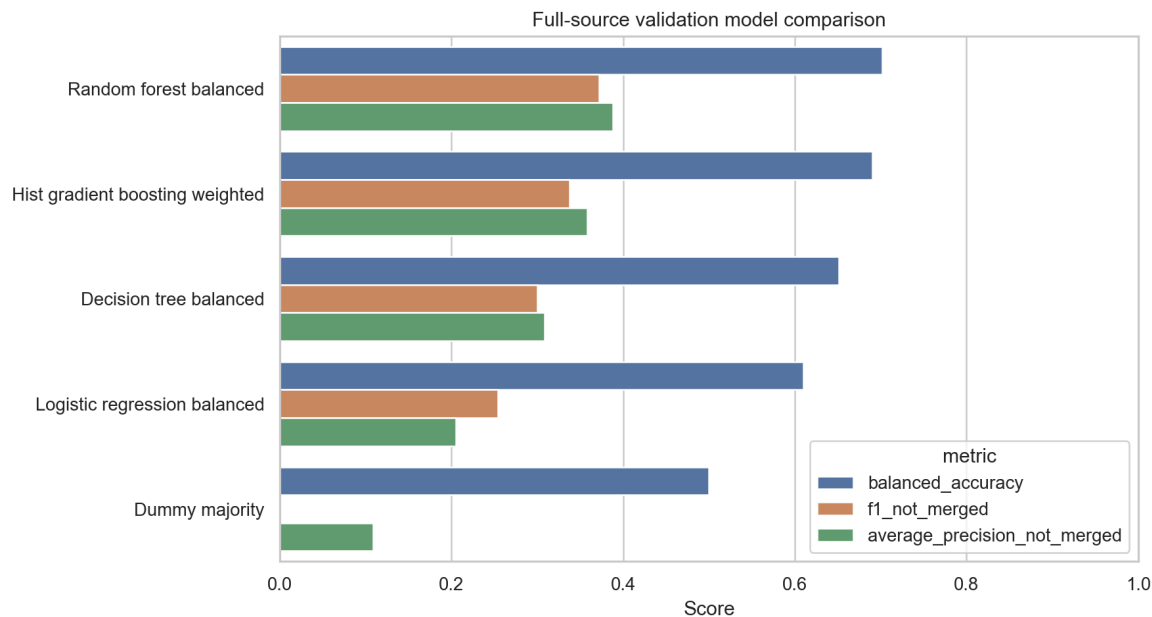
EDA supports the modeling story: class imbalance, contributor context, project scale, and repeated entities all matter.

Model Design and Algorithm Comprehension

Method	Why used	Limitation
Dummy majority	exposes class-imbalance trap	detects no not-merged PRs
Logistic regression	interpretable weighted linear baseline	weak on interactions
Decision tree	simple non-linear rules	unstable if unconstrained
Random forest	averages balanced trees	importance is not causal
Histogram gradient boosting	boosted supervised comparator	close, but lower primary F1
K-means + PCA	unsupervised profile analysis	interpretive, not predictive

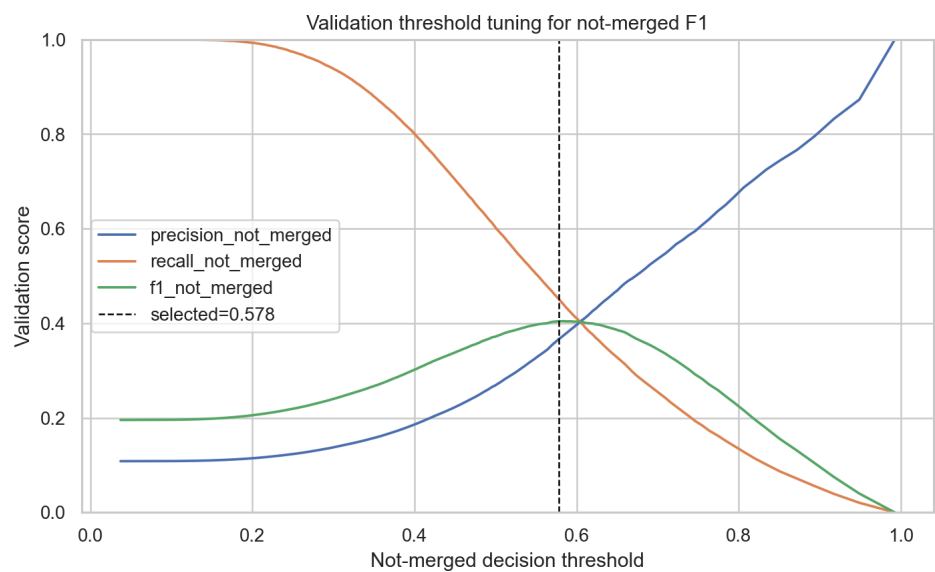
Selection rule: maximize **not-merged F1** on internal validation, then retrain once on all training rows.

Validation Comparison



Model	Balanced acc.	Not-merged recall	Not-merged F1	ROC-AUC
Random forest balanced	0.702	0.602	0.372	0.778
Histogram gradient boosting	0.690	0.644	0.337	0.759
Logistic regression balanced	0.610	0.595	0.254	0.655
Dummy majority	0.500	0.000	0.000	0.500

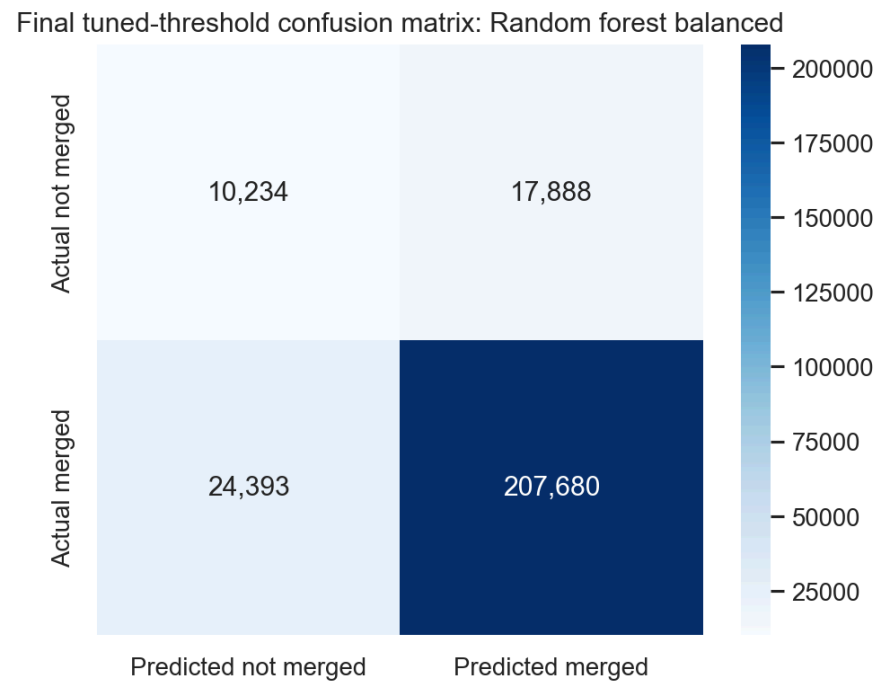
Threshold Tuning and Final Test



Metric	Default	Tuned
Accuracy	0.756	0.838
Balanced acc.	0.650	0.629
Precision, not merged	0.225	0.296
Recall, not merged	0.515	0.364
F1, not merged	0.314	0.326
Avg precision, not merged	0.301	0.301

Tuned threshold: 0.578 , selected on validation predictions only.

Final Confusion Matrix



Tuned-threshold counts:

Correct not merged	Missed not merged	False not merged	Correct merged
10,234	17,888	24,393	207,680

The model finds signal, but it is not reliable enough for automated decisions.

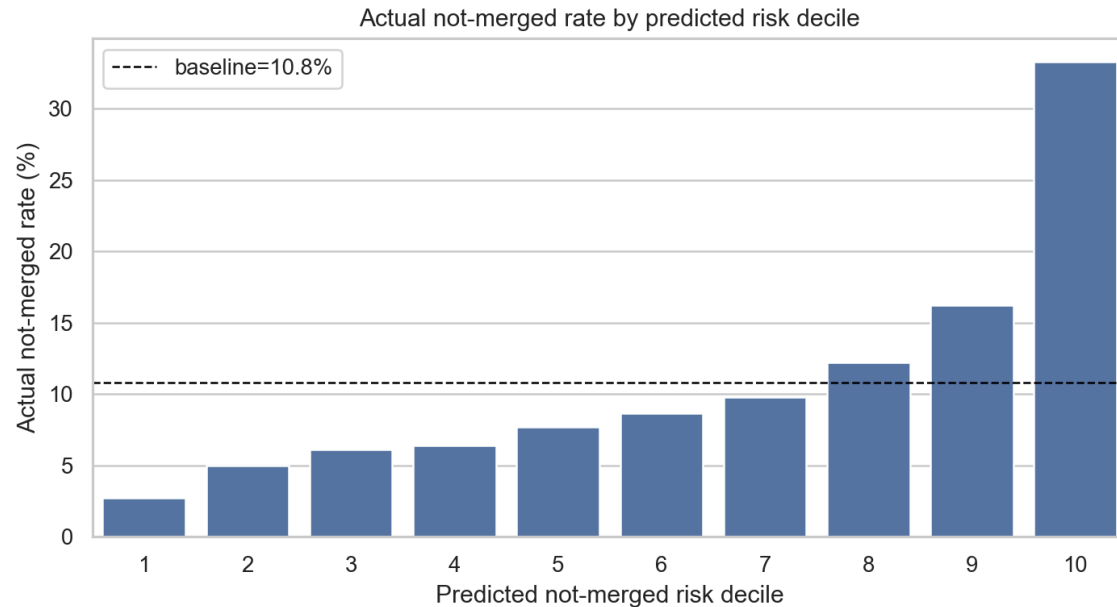
Prediction-Time Contracts

Contract	What is allowed	Test F1	Test AP
T0 creation	history, project snapshots, description	0.311	0.302
T1 submitted diff	T0 + initial diff/test inclusion	0.314	0.301
T2 review process	T1 + comments, CI progress, integrator context	0.436	0.479

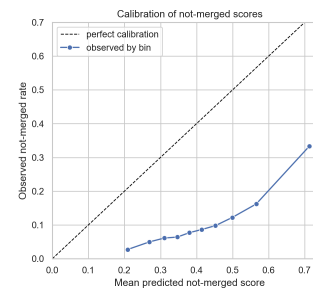
This is the upgraded answer: stronger metrics require later review-process information.

The early model is still moderate; the late model shows where extra signal enters.

Risk Ranking, Not Certainty

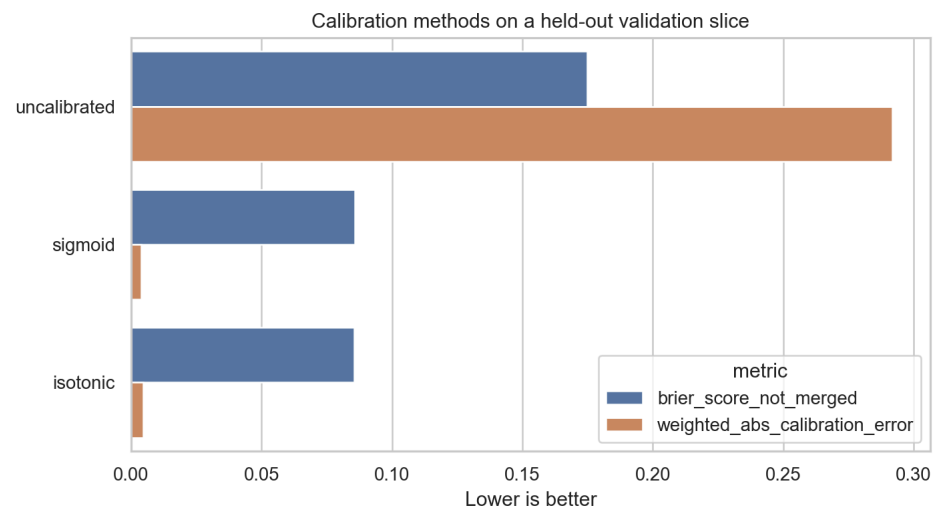


- The score is most defensible as **relative risk**, not a literal probability.
- High predicted-risk bands should contain more not-merged PRs than the baseline.
- Calibration is checked separately so the project does not overclaim probability quality.



Professor-facing claim: the model can rank risk better than chance, but it cannot make reliable merge decisions.

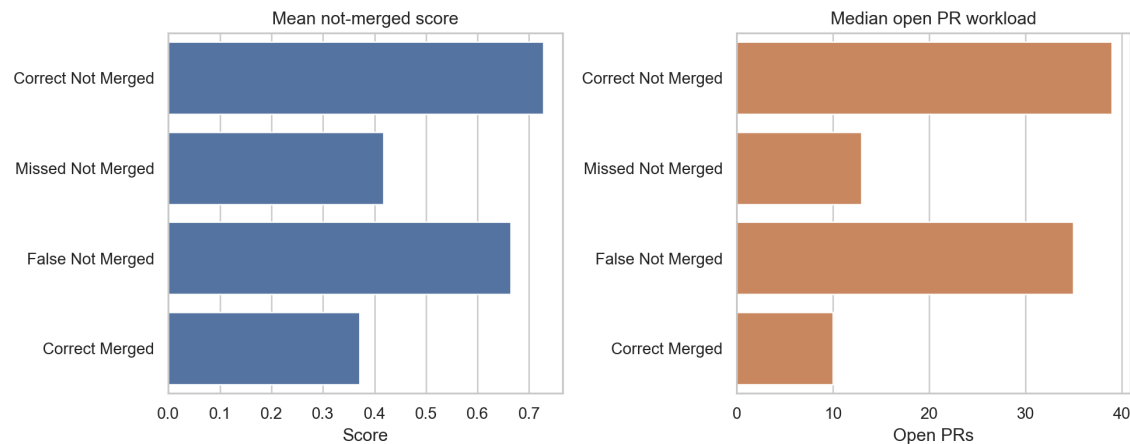
Calibration Upgrade



Method	Brier	Cal. error	AP	F1
Uncalibrated	0.175	0.292	0.326	0.345
Sigmoid	0.086	0.004	0.326	0.123
Isotonic	0.086	0.005	0.314	0.136

Calibration fixes probability honesty. It does not optimize F1, and calibrated default-threshold F1 drops because the not-merged base rate is only about 11%.

Where the Model Fails



- **Missed not-merged PRs:** not-merged cases that look too similar to merged PRs under the available features.
- **False not-merged predictions:** accepted PRs in contexts that look risky, especially busier/larger repositories.
- This is why the final answer stays bounded to **moderate association**.

Error analysis strengthens the conclusion because it explains failure modes instead of hiding them.

Split Overlap and Validation-Test Gap

Entity	Test unique overlap	Test rows with seen entity
PR id	0.00%	0.00%
Project	94.91%	97.85%
Creator	69.39%	84.09%

Evaluation	Threshold	Not-merged F1	Average precision
Validation	default	0.372	0.388
Validation	tuned	0.405	0.388
Official test	default	0.314	0.301
Official test	tuned	0.326	0.301

The official holdout has clean PR ids, but it is not an unseen-project or unseen-creator test.

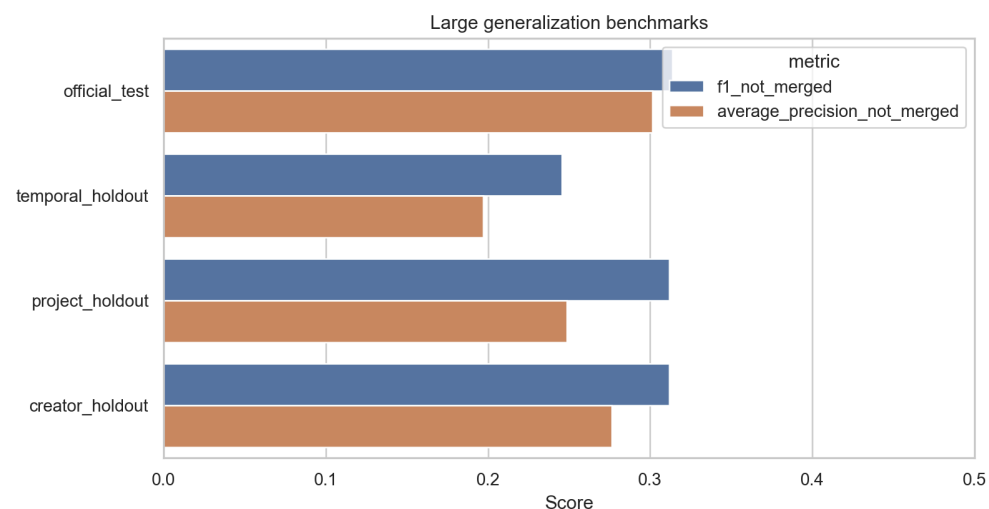
Generalization Stress Tests

Sampled stress tests, each using the same headline feature policy:

Stress test	Balanced acc.	Precision	Recall	F1
Random stratified	0.684	0.258	0.566	0.354
Temporal last 25%	0.610	0.196	0.342	0.250
Project-group holdout	0.593	0.185	0.381	0.249
Creator-group holdout	0.641	0.227	0.479	0.308

Stress tests are the external-validity reality check: the signal weakens for later or unseen-project-like validation.

Larger Generalization Benchmarks



Benchmark	Train rows	Eval rows	F1	AP
Official test	1,045,883	260,195	0.314	0.301
Temporal holdout	375,000	125,000	0.246	0.197
Project holdout	372,830	127,170	0.312	0.249
Creator holdout	375,150	124,850	0.312	0.276

The sampled and larger holdouts are different diagnostics, so the exact values need not match.

The shared conclusion is stable: the official split is useful, but stricter holdouts change the story.

Threshold and Model Robustness

What this slide defends:

Concern	Evidence
One lucky threshold	repeated sampled folds select thresholds from 0.542 to 0.564
One lucky model split	model families are compared under random, temporal, project, and creator splits
Random Forest vs HGB	paired validation delta: +0.035 F1, 95% CI +0.033 to +0.037
Absolute winner claim	not made; HGB narrowly leads the project-group diagnostic

The submitted model remains Random Forest because the selection rule was declared before test scoring.

Uncertainty: Row vs Project-Cluster Bootstrap

Tuned-threshold 95% intervals:

Metric	Estimate	Row bootstrap	Project-cluster bootstrap
Precision, not merged	0.296	0.290 to 0.300	0.265 to 0.327
Recall, not merged	0.364	0.358 to 0.370	0.323 to 0.408
F1, not merged	0.326	0.321 to 0.331	0.296 to 0.360
Balanced accuracy	0.629	0.626 to 0.632	0.611 to 0.650

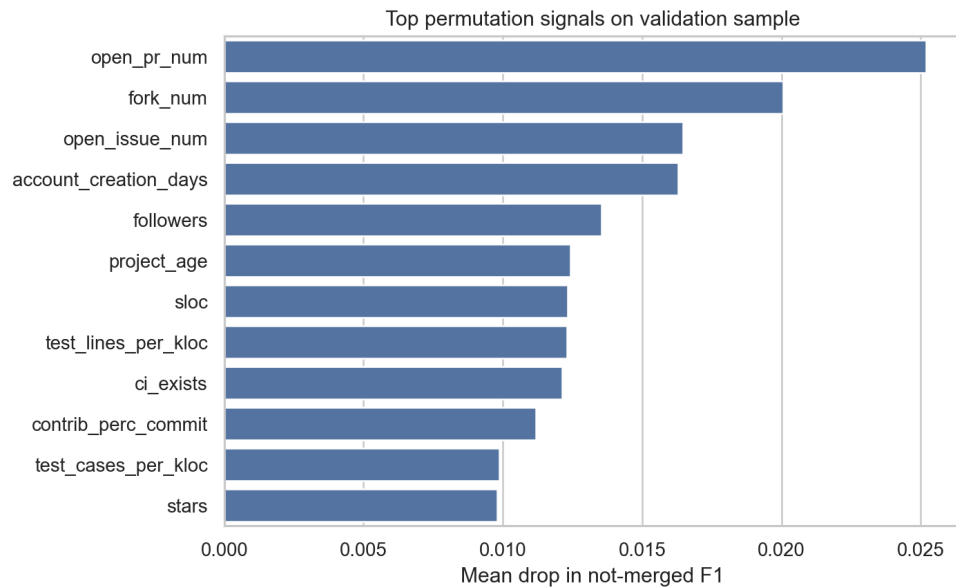
Project-cluster intervals are wider because PRs are correlated within repositories.

Feature-Policy Sensitivity

Feature policy	Validation F1	Test F1	Interpretation
Ultra-conservative	0.310	0.273	strictest feature contract
Headline leakage-safer	0.372	0.314	submitted result
Integrator-assumed	0.424	0.373	stronger timing assumption
Extended timing-assumed	0.375	0.315	little gain over headline

The strongest-looking feature policy is not the headline model because it depends on stronger availability assumptions.

Explanation Evidence



Top permutation signals:

- open_pr_num
- fork_num
- open_issue_num
- account_creation_days
- followers
- project_age
- sloc
- test_lines_per_kloc
- ci_exists
- contrib_perc_commit

Permutation importance and ablation are explanation diagnostics, not causal evidence.

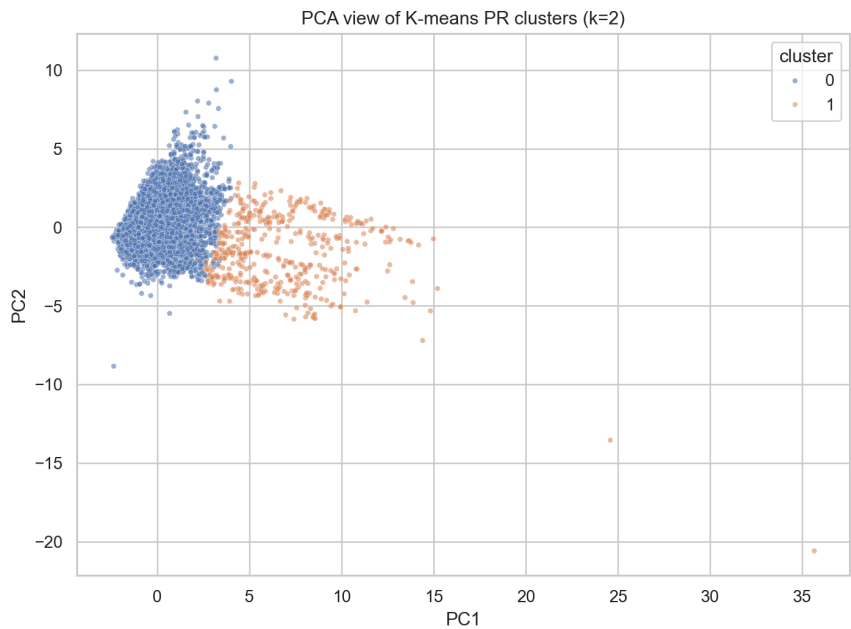
Feature-Family Ablation

Validation sample ablation against the full headline feature policy:

Removed family	Not-merged F1	Delta F1	Avg precision delta
None	0.354	0.000	0.000
Contributor history	0.335	-0.019	-0.020
Project context	0.324	-0.030	-0.040
PR scope	0.354	-0.001	+0.007
Testing/CI context	0.342	-0.012	-0.014
Language/calendar	0.354	0.000	-0.003

Project context and contributor history are the most important feature families in this diagnostic.

Unsupervised PR Profiles



Cluster	Size	Merge rate	Not-merged rate	Profile
0	67,183	89.57%	10.43%	lower not-merged / larger-change / smaller-project
1	2,817	79.45%	20.55%	higher not-merged / smaller-change / larger-project

K-means/PCA is a profile analysis, not a second classifier.

Assignment Coverage

Criterion	Evidence
Problem clarity	question, target, scope, feature-availability contract
EDA	concrete findings, imbalance, concentration, class-conditioned patterns
Empirical study	contracts, baselines, final test, calibration, holdouts, paired deltas
Algorithm comprehension	role and limitation table for every method
Data characteristics	missingness, skew/outliers, leakage timing, split overlap, comment join limit
Findings/report/slides	final notebook, report PDF, generated metrics, discussion deck

Full evidence map: `deliverables/final/assignment_coverage.csv`

Threats to Validity

- Feature timing is the main risk; risky fields are excluded or sensitivity-only.
- The official split has heavy project and creator overlap.
- Review-process features improve metrics but are not early-prediction features.
- The comment dataset is profiled, not force-joined, because local join keys do not match.
- Random Forest explanations are predictive, not causal.
- The model is not ready for deployment or automated PR decisions.

Professor-facing bottom line: the analysis is stronger because it admits these limits directly.

Final Answer

The project does answer the chosen question.

Signal depends on when prediction is made.

- Early T1 submitted-diff test F1: **0.314** default, **0.326** tuned.
- T2 review-process test F1: **0.436**, but it uses later information.
- Calibration can make probabilities honest; it does not make decisions reliable.
- Larger holdouts show the official split is not the whole generalization story.
- The right conclusion is empirical association, not causality or automation.

Q&A framing: stronger metrics come from later information, not from pretending the early model is better.